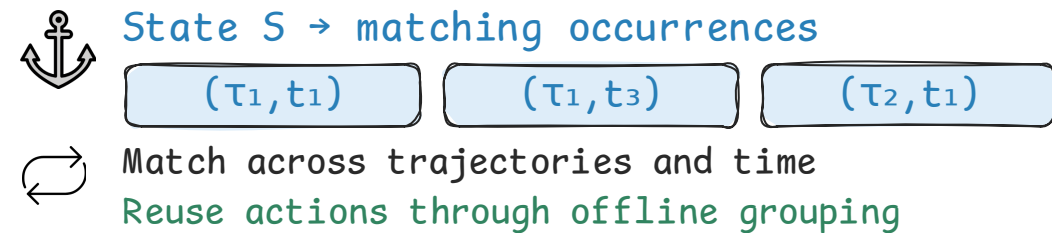
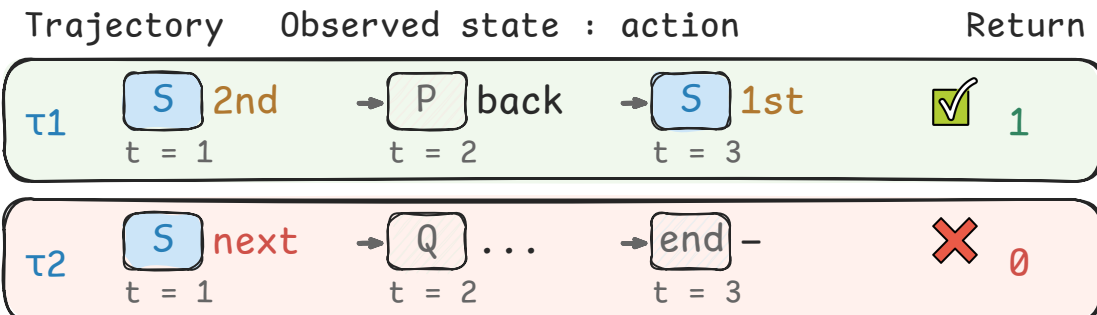
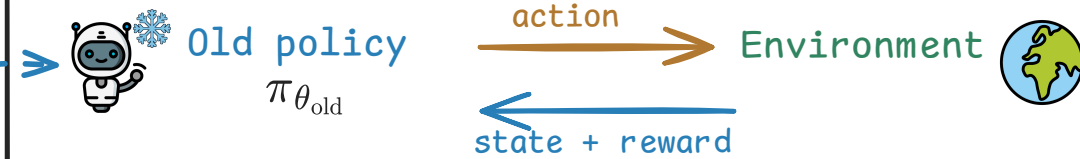


## Collect a shared trajectory group

 Same task  $\times$  + initial state  $S$   
 N copies of the environment



 Clipped action objective + reference KL

  $\rho$ : current / old action probability

## Episode &amp; anchor-state comparisons

 Episode relative advantage

$$E_i = \sum_t r_t^{(i)}, \quad A_i^E = \frac{E_i - \mu_E}{F_E}$$

Compare total returns across trajectories

 Step relative advantage

$$Q_{i,t} = \sum_{k=t}^{T_i} \gamma^{k-t} r_k^{(i)}$$

Use discounted future returns

$$A_{i,t}^S = \frac{Q_{i,t} - \mu_{\mathcal{I}(S)}}{F_{\mathcal{I}(S)}}$$

Compare actions within the same state  
 Normalization F: group std or 1

## Fuse credit &amp; update

 Global + local

$$A_{i,t} = A_i^E + \omega A_{i,t}^S$$

$\omega$ : global / local balance  
 Different credit for each step

 Trainable actor  
 $\pi_{\theta}$  Clipped update

 Reference policy  
 $\pi_{\text{ref}}$  KL anchor

 Action-specific credit  
 No extra rollout or critic

$$\hat{J} = \frac{1}{NT} \sum_{i,t} \min(\rho_{i,t} A_{i,t}, \text{clip}(\rho_{i,t}, 1 - \epsilon, 1 + \epsilon) A_{i,t}) - \beta D_{\text{KL}}(\pi_{\theta} \| \pi_{\text{ref}})$$

KL: keep the actor close to the reference